

Speed, Time and Distance: Formulas, Relationship, Applications with Examples

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The basic concept of speed, time and distance is the relation between three variables. One-two word problems are proposed based on speed, time and distance with variation but students must also retain themselves ready to see questions on data sufficiency and data interpretation that are based on TDS (i.e time, distance and speed) topic. Questions related to speed, distance and time include various categories such as straight line, relative motion, circular motion, trains, boats, clocks, races, etc.

What are Speed, Time and Distance?

Speed of a body is the distance covered by the body per unit time i.e. $\text{Speed} = \text{Distance}/\text{Time}$.

Speed: Speed is the rate at which a particular distance is covered by an object in motion.

Time: Time is an interval separating two events.

Distance: Distance is the extent of space between two points.

Units of Speed Time and Distance

Each of the speed, distance and time can be represented in different units:

- Time can be generally expressed in terms of seconds(s), minutes (min) and hours (hr).
- Whereas the distance is generally expressed in meters (m), kilometres (km), centimetres, miles, feet, etc.
- Speed is commonly expressed in m/s, km/hr.

Example: If the distance is given in km and time in hr then as per the formula:

$\text{Speed} = \text{Distance} / \text{Time}$; the [unit of speed](#) will become km/ hr.

Relationship Between Speed, Time and Distance

Now that we are well aware of the definition of speed, distance and time let us understand the relationship between them. It is said that an object attains motion or movement when it changes its position with respect to any external stationary point. Speed, Time and Distance are the three variables that represent the mathematical model of motion as, $s \times t = d$.

- Time is directly proportional to distance. It means that speed remains constant, if we have two vehicles moving two distances for two different time duration then the time is directly proportional to the distance.

- Speed is [directly proportional](#) to distance. It means that time remains constant if we have two vehicles moving two distances at two different speeds respectively.
- Speed is [Inversely Proportional](#) to time. It means that distance remains constant if we have two vehicles moving at two different speeds and taking times respectively.

Formula for speed calculation is **Speed = Distance/Time**

This shows us how slow or fast a target moves. It represents the distance covered divided by the time needed to cover the distance.

Speed is directly proportional to the given [distance](#) and inversely proportional to the proposed time. Hence,

Distance = Speed x Time and

Time = Distance / Speed since as the speed grows the time needed will decrease and vice versa.

Conversion of Speed, Time and Distance

The conversion of speed, time and distance into various units is given below:

- To convert a given data from km/hour to m/sec, we multiply by 5/18. As 1 km/hour = 5/18 m/sec.
- To convert a given data m/sec to km/hour, we multiply by 18/5. As 1 m/sec = 18/5 km/hour = 3.6 km/hour.

In terms of formula, we can list it as:

$$x \text{ km/hr} = x \times \frac{5}{18} \text{ m/sec}$$

$$x \text{ m/sec} = x \times \frac{18}{5} \text{ km/hr}$$

Similarly, some other conversions are given below:

- 1 km/hr = 5/8 miles/hour
- 1 yard = 3 feet
- 1 kilometer = 1000 meters
- 1 mile = 1.609 kilometer
- 1 hour = 60 minutes = 3600 seconds
- 1 mile = 1760 yards
- 1 yard = 3 feet
- 1 mile = 5280 feet

Applications of Speed, Time and Distance

Some of the major applications of speed, time and distance are given below:

Average Speed: The average speed is determined by the formula = (Total distance travelled)/(Total time taken)

$$\text{Average speed} = \frac{d_1 + d_2 + d_3 \cdots d_n}{t_1 + t_2 + t_3 \cdots t_n}$$

Sample 1 – When the distance travelled is constant and two speed is given then:

$$\text{Average speed} = \frac{2xy}{x+y}$$

where x and y are the two speeds at which the corresponding distance has been reached.

Sample 2 – When the time taken is constant average speed is calculated by the formula:

$$\text{Average speed} = \frac{(x+y)}{2}$$

where x and y are the two speeds at which we covered the distance for an identical time.

Example: An individual drives from one place to another at 40 km/hr and returns at 160 km/hr. If the complete time needed is 5 hours, then obtain the distance.

Solutions: Here the distance is fixed, so the time taken will be inversely proportional to the speed. The ratio of speed is given as 40:160, i.e. 1:4.

Therefore the ratio of time taken will be 4:1.

Total time is practised = 5 hours; therefore the time taken while travelling is 4 hours and returning is 1 hour.

Hence, distance = 40x 4 = 160 km.

If the first part of any given distance is covered at a rate of v_1 in time t_1 and the second part of the distance is covered at a rate v_2 in time t_2 then the average speed is given by the formula:

$$\text{Average speed} = \frac{(v_1 t_1 + v_2 t_2)}{t_1 + t_2}$$

Relative Speed: As the name suggests the idea is about the relative speed between two or more things. The basic concept of relative speed is that the speed gets combined in the case of objects moving in the opposite direction to one another and the speed gets subtracted for the case when objects are moving in an identical direction.

For example, if two passenger trains are moving in the opposite direction with a speed of X km per hour and Y kilometre per hour respectively. Then their relative speed is given by the formula:

$$\text{Relative speed} = X + Y$$

On the other hand, if the two trains are travelling in the same direction with the speed of X km per hour and Y kilometre per hour respectively. Then their relative speed is given by the formula:

$$\text{Relative speed} = X - Y$$

For the first case time taken by the train in passing each other is given by the formula:

Relative speed = X + Y

$$\text{Time taken} = \frac{L_1 + L_2}{X + Y}$$

For the second case, the time taken by the trains in crossing each other is given by the formula:

Relative speed = X - Y

$$\text{Time taken} = \frac{L_1 + L_2}{X - Y}$$

Here L_1 , L_2 are the lengths of the trains respectively.

Inverse Proportionality of Speed & Time: Speed is said to be inversely proportional to time when the distance is fixed. In mathematical format, S is inversely proportional to 1/T when D is constant. For such a case if the speeds are in the ratio m:n then the time taken will be in the ratio n:m.

There are two approaches to solving questions:

- Applying Inverse Proportionality
- Applying Constant [Product Rule](#)

Example: After moving 100km, a train meets with an accident and travels at $\left(\frac{3}{4}\right)^{th}$ of the normal speed and reaches 55 min late. Had the accident occurred 20 km further on it would have arrived 45 min delayed. Obtain the usual Speed?

Solutions: Applying Inverse Proportionality Method

Here there are 2 cases

Case 1: accident happens at 100 km

Case 2: accident happens at 120 km

The difference between the two incidents is only for the 20 km between 100 km and 120 km. The time difference of 10 minutes is just due to these 20 km.

In case 1, 20 km between 100 km and 120 km is covered at $\left(\frac{3}{4}\right)^{th}$ speed.

In case 2, 20 km between 100 km and 120 km is reached at the usual speed.

So the usual time “t” taken to cover 20 km, can be found as follows. $\frac{4}{3}t - t = 10 \text{ mins} \Rightarrow t = 30 \text{ mins}$, d = 20 km

so the usual speed = $20/30 \text{ min} = 20/0.5 = 40 \text{ km/hr}$

Using Constant Product Rule Method: Let the actual time taken be equal to T.

There is a $(1/4)$ th reduction in speed, this will result in a $(1/3)$ rd increase in time taken as speed and time are inversely proportional to one another.

A $1/x$ increment in one of the parameters will result in a $1/(x+1)$ reduction in the other parameter if the parameters are inversely proportional.

The delay due to this reduction is 10 minutes

Thus $1/3 T = 10$ and $T = 30$ minutes or 0.5 hour

Also, Distance = 20 km

Thus Speed = 40 kmph

Meeting Point Question

If two individuals travel from two locations P and Q towards each other, and they meet at point X. Then the total distance traversed by them at the meeting will be PQ. The time taken by both of them to meet will be identical.

As the time is constant, the distances PX and QX will be in the ratio of their speed. Assume that the distance between P and Q is d.

If two individuals are stepping towards each other from P and Q respectively, when they meet for the first time, they collectively cover a distance “d”. When they meet each other for the second time, they mutually cover a distance “3d”. Similarly, when they meet for the third time, they unitedly cover a distance of “5d” and the process goes on.

Take an example to understand the concept:

Example: Ankit and Arnav have to travel from Delhi to Hyderabad in their respective vehicles. Ankit is driving at 80 kmph while Arnav is operating at 120 kmph. Obtain the time taken by Arnav to reach Hyderabad if Ankit takes 9 hrs.

Solutions: As we can recognise that the distance covered is fixed in both cases, the time taken will be inversely proportional to the speed. In the given question, the speed of Ankit and Arnav is in a ratio of 80: 120 or 2:3.

Therefore the ratio of the time taken by Ankit to that taken by Arnav will be in the ratio 3:2. Hence if Ankit takes 9 hrs, Arnav will take 6 hrs.

Learn about [Time and Work](#)

Speed, Time and Distance Formulas

Some important speed, distance and time formulas are given below:

Terms	Formula
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Speed	$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$
Time	$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$
Distance	$D = (\text{Speed} \times \text{Time})$
Average Speed	$\text{Average Speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$
Average Speed (when the distance travelled is constant)	$\frac{2xy}{x+y}$
Relative speed (If two trains are moving in the opposite direction)	Relative speed = $X + Y$ Time taken = $\frac{L_1 + L_2}{X + Y}$ Here L_1, L_2 are the lengths of the trains.
Relative speed (If two trains are moving in the same direction)	Relative speed = $X - Y$ Time taken = $\frac{L_1 + L_2}{X - Y}$ Here L_1, L_2 are the lengths of the trains.

Some additional formulas of speed, distance and time are:

- If the ratio of the speeds of P and Q is p: q, then the ratio of the times used by them to reach the same distance is $1/p:1/q$ or q: p.
- If two individuals or automobiles or trains start at the exact time in the opposing direction from two points say A and B and after crossing each other they take time a and b respectively to finish the journey then the speed ratio is provided by the formula:

$$\frac{\text{Speed of first}}{\text{Speed of second}} = \sqrt{\frac{b}{a}}$$
- If two individuals with two different speeds x and y cover the same distance and travel in opposite directions. Where the total time is given and distance is asked then the formula is:

$$\text{Distance} = \frac{xy}{x+y} \times \text{Total Time}$$

Types of Questions from Speed, Time and Distance

There are some specific types of questions from Speed, time and distance that usually come in exams. Some of the important types of questions from speed, distance and time are as follows.

(a) Problems related to Trains

Please note that, in the case of the train problems, the distance to be covered when crossing an object is equal to,
Distance to be covered = Length of train + Length of object.

Remember that, in case the object under consideration is a pole or a person or a point, we can consider them to be point objects with zero length. It means that we will not consider the lengths of these objects. However, if the object under consideration is a platform (non-point object), then its length will be added to the formula of the distance to be covered.

(b) Boats and Streams

In such problems, boats travel either in the direction of stream or in the opposite direction of stream. The direction of boat along the stream is called downstream and the direction of boat against the stream is called upstream.

If the speed of a boat in still water is u km/hr and the speed of the stream is v km/hr, then:

1) Speed downstream = $(u + v)$ km/hr

2) Speed upstream = $(u - v)$ km/hr

Learn about [Ratio and Proportion](#) and [Mixture and Alligation](#)

Tips and Tricks to Solve Questions based on Speed, Time and Distance

Students can find different tips and tricks below for solving the questions based on speed, time and distance.

Tip 1: Relative speed is defined as the speed of a moving body with respect to another body. The possible cases of relative motion are, same direction, when two bodies are moving in the same direction, the relative speed is the difference between their speeds and is always expressed as a positive value. On the other hand, the opposite direction is when two bodies are moving in the opposite direction, the relative speed is the sum of their speeds.

Tip 2: Average speed = Total Distance / Total Time

Tip 3: When train crossing a moving body,

When a train passes a moving man/point object, the distance travelled by train while passing it will be equal to the length of the train and relative speed will be taken as

1) If both are moving in same direction then relative speed = Difference of both speeds

2) If both are moving in opposite direction then relative speed = Addition of both speeds

Tip # 4: Train Passing a long object or platform, when a train passes a platform or a long object, the distance travelled by the train, while crossing that object will be equal to the sum of the length of the train and length of that object.

Tip # 5: Train passing a man or point object, when a train passes a man/object, the distance travelled by the train while passing that object, will be equal to the length of the train.

Solved Examples of Speed, Time and Distance

Some of the solved questions regarding the topic for more practice are as follows:

Example 1: The speed of three cars are in the ratio 5 : 4 : 6. The ratio between the time taken by them to travel the same distance is

Solution: Ratio of time taken = $\frac{1}{5} : \frac{1}{4} : \frac{1}{6} = 12 : 15 : 10$

Example 2: A truck covers a distance of 1200 km in 40 hours. What is the average speed of the truck?

Solution: Average speed = Total distance travelled/Total time taken

$$\Rightarrow \text{Average speed} = 1200/40$$

$$\therefore \text{Average speed} = 30 \text{ km/hr}$$

Example 3: A man travelled 12 km at a speed of 4 km/h and further 10 km at a speed of 5 km/hr. What was his average speed?

Solution: Total time taken = Time taken at a speed of 4 km/h + Time taken at a speed of 5 km/h

$$\Rightarrow 12/4 + 10/5 = 5 \text{ hours } [\because \text{Time} = \text{Distance}/\text{Speed}] \text{ Average speed} = \text{Total distance}/\text{Total time}$$

$$\Rightarrow (12 + 10) / 5 = 22/5 = 4.4 \text{ km/h}$$

Example 4: Rahul goes Delhi to Pune at a speed of 50 km/h and comes back at a speed of 75 km/h. Find his average speed of the journey.

Solution: Distance is same both cases

$$\Rightarrow \text{Required average speed} = (2 \times 50 \times 75)/(50 + 75) = 7500/125 = 60 \text{ km/hr}$$

Example 5: Determine the length of train A if it crosses a pole at 60km/h in 30 sec.

Solution: Given, speed of the train = 60 km/h

$$\Rightarrow \text{Speed} = 60 \times 5/18 \text{ m/s} = 50/3 \text{ m/s}$$

Given, time taken by train A to cross the pole = 30 s

The distance covered in crossing the pole will be equal to the length of the train.

$$\Rightarrow \text{Distance} = \text{Speed} \times \text{Time}$$

$$\Rightarrow \text{Distance} = 50/3 \times 30 = 500 \text{ m}$$

Example 6: A 150 m long train crosses a 270 m long platform in 15 sec. How much time will it take to cross a platform of 186 m?

Solution: In crossing a 270 m long platform,

Total distance covered by train = $150 + 270 = 420$ m

Speed of train = total distance covered/time taken = $420/15 = 28$ m/sec In crossing a 186 m long platform,

Total distance covered by train = $150 + 186 = 336$ m

$\therefore$ Time taken by train = distance covered/speed of train = $336/28 = 12$ sec.

Example 7: Two trains are moving in the same direction at speeds of 43 km/h and 51 km/h respectively. The time taken by the faster train to cross a man sitting in the slower train is 72 seconds. What is the length (in metres) of the faster train?

Solution: Given: The speed of 2 trains = 43 km/hr and 51 km/hr Relative velocity of both trains = $(51 - 43)$ km/hr = 8 km/hr Relative velocity in m/s = $8 \times (5/18)$ m/s

$\Rightarrow$ Distance covered by the train in 72 sec = $8 \times (5/18) \times 72 = 160$ Hence, the length of faster train = 160 m

Example 8: How long will a train 100m long travelling at 72km/h take to overtake another train 200m long travelling at 54km/h in the same direction?

Solution: Relative speed = $72 - 54$ km/h (as both are travelling in the same direction)

= 18 km/hr = $18 \times 10/36$ m/s = 5 m/s

Also, distance covered by the train to overtake the train = 100 m + 200 m = 300 m Hence,

Time taken = distance/speed = $300/5 = 60$ sec

Example 9: A boat takes 40 minutes to travel 20 km downstream. If the speed of the stream is 2.5 km/hr, how much more time will it take to return back?

Solution: Time taken downstream = 40 min = $40/60 = 2/3$ hrs. Downstream speed = $20 / (2/3) = 30$ km/hr.

As we know, speed of stream = $1/2 \times (\text{Downstream speed} - \text{Upstream speed})$

$\Rightarrow$ Upstream speed = $30 - 2 \times 2.5 = 30 - 5 = 25$ km/hr.

Time taken to return back = $20/25 = 0.8$ hrs. = $0.8 \times 60 = 48$ min.

$\therefore$ The boat will take = $48 - 40 = 8$ min. more to return back.

We hope you found this article regarding Speed, Time and Distance was informative and helpful, and please do not hesitate to contact us for any doubts or queries regarding the same. You can also download the [Testbook App](#), which is absolutely free and start preparing for any government competitive examination by taking the mock tests before the examination to boost your preparation.

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FAQs on Speed, Time and Distance

Q.1 What is speed, time and distance?

Ans.1 Speed points us to how slowly or quickly an object moves and time refers to an interval dividing two events whereas distance as the name suggests points towards the extent of space between two points.

Q.2 What is the formula of speed, distance, and time?

Ans.2 The formula of speed, time, and distance are as follows:

Speed = distance/time

Distance = speed × time

Time = distance/speed

Q.3 What is the relationship between speed, distance and time?

Ans.3 The relationship between speed, distance and time are:

Speed and time are directly proportional to distance. Also, speed is inversely proportional to time.

Q.4 Is speed directly proportional to distance?

Ans.4 Consider the basic formula: Speed= distance/time.

The formula states that speed is directly proportional to distance.

Q.5 What is relative speed?

Ans.5 The basic concept of relative speed is that the speed gets combined in the case of objects moving in the opposite direction to one another and the speed gets subtracted for the case when objects are moving in an identical direction.

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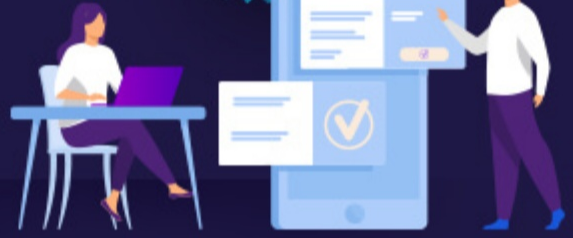


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